

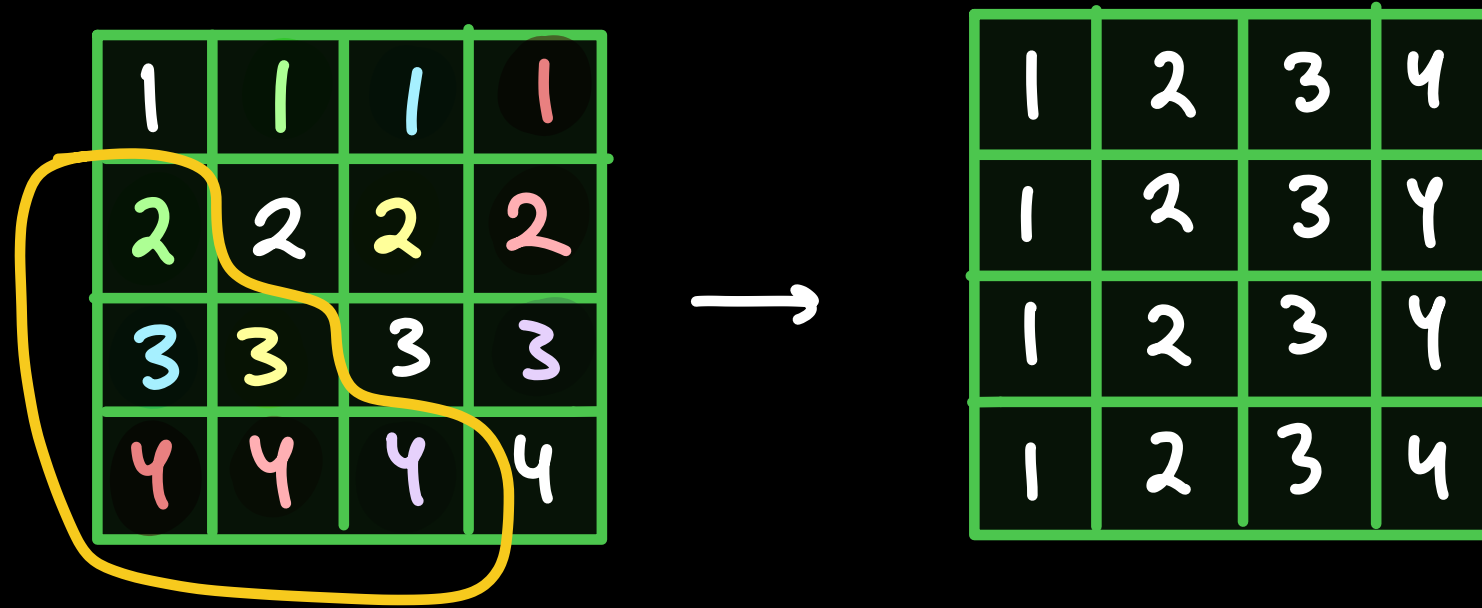


HW · Print matrix in Snake Pattern

6	0	9	2	3
5	2	6	0	9
8	4	8	1	7
4	1	7	3	5

Ques: Transpose of Matrix

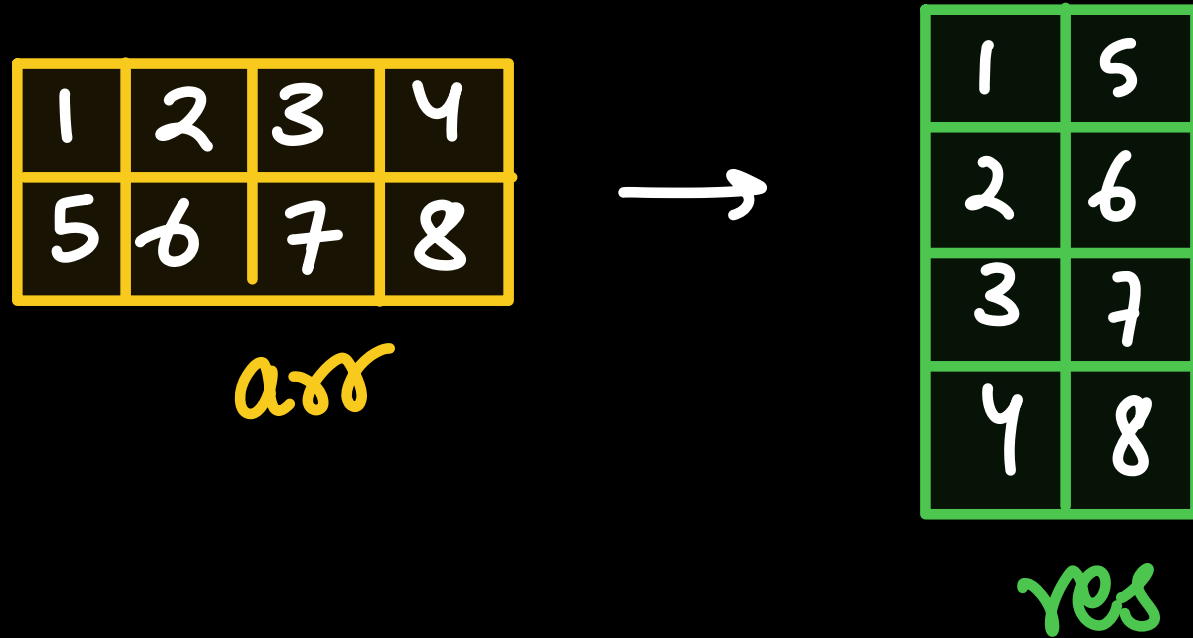
(hint: swap)  Earn ₹s



$\text{swap}(\text{arr}[i][j], \text{arr}[j][i])$

GFG

HW : Transpose of Matrix



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Ques: 48. Rotate Image

(Hint: Transpose)  Earners

1	2	3
4	5	6
7	8	9

1	4	7
2	5	8
3	6	9

7	4	1
8	5	2
9	6	3

transpose

reverse
each
row

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HW: Rotate by 90 degree

[[0, 1, 2],
[3, 4, 5],
[6, 7, 8]]



[[2, 5, 8],
[1, 4, 7],
[0, 3, 6]]

GFG

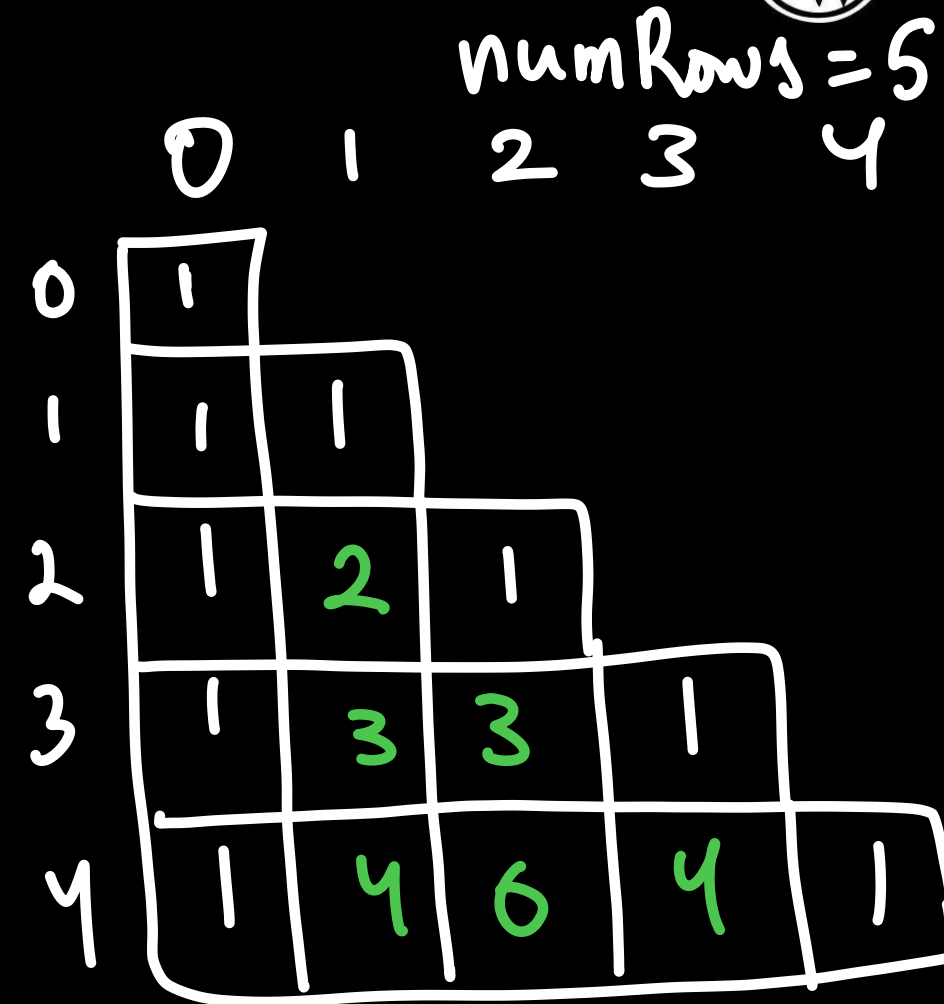
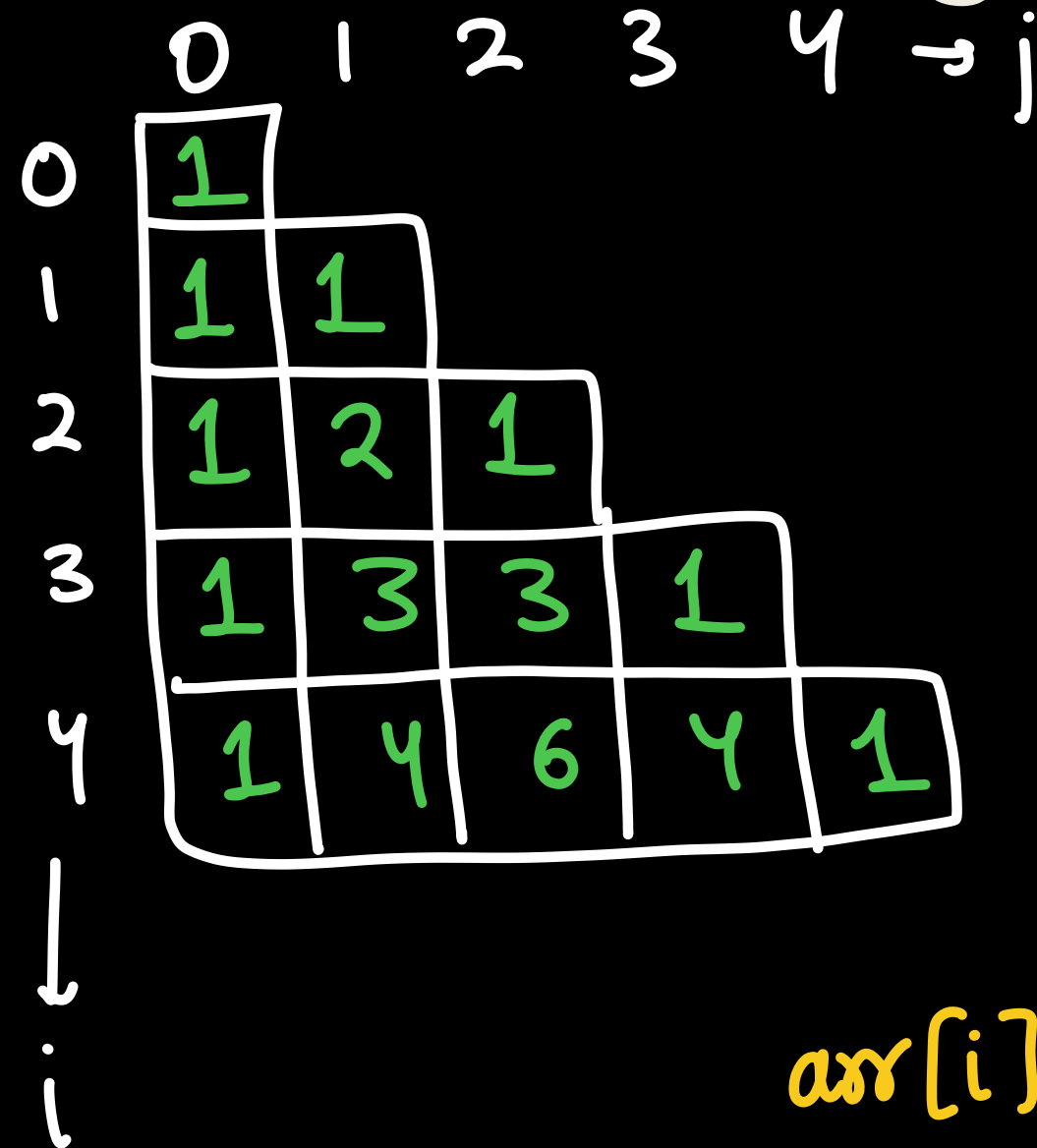
2D Vector

	0	1	2	3
0	3	7	0	3
1	4			
2	9	1	4	
3	2	3		

arr

$$\text{arr} = \{ \{3, 7, 0, 3\}, \{4\}, \{9, 1, 4\}, \{2, 3\} \}$$

Ques: Pascal Triangle



$$arr[i][j] = arr[i-1][j-1] + arr[i-1][j]$$

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Ques: Multiply Matrices

$$\begin{array}{c}
 \begin{array}{cc}
 & \begin{array}{cc} 0 & 1 \end{array} \\
 \begin{array}{c} 0 \\ 1 \end{array} & \begin{array}{|cc|} \hline 1 & 2 \\ \hline 1 & 1 \\ \hline \end{array} \\
 & a
 \end{array}
 \times
 \begin{array}{c}
 \begin{array}{cc}
 & \begin{array}{cc} 0 & 1 \end{array} \\
 \begin{array}{c} 0 \\ 1 \end{array} & \begin{array}{|cc|} \hline 2 & 0 \\ \hline -1 & 3 \\ \hline \end{array} \\
 & b
 \end{array}
 =
 \begin{array}{c}
 \begin{array}{cc}
 & \begin{array}{cc} 0 & 1 \end{array} \\
 \begin{array}{c} 0 \\ 1 \end{array} & \begin{array}{|cc|} \hline 0 & 6 \\ \hline 1 & 3 \\ \hline \end{array} \\
 & \text{res}
 \end{array}
 \end{array}$$

$$\text{res}[0][0] = 1 \times 2 + 2 \times -1 = 0$$

$$1 \times 0 + 2 \times 3 = 6$$

$$1 \times 2 + 1 \times -1 = 1$$

$$1 \times 0 + 1 \times 3 = 3$$



Ques: Multiply Matrices

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

a

	0	1	2
0	1	-2	4
1	2	7	2
2	3	8	0

b

=

	0	1	2
0			
1			
2			

res

$$TC = O(n^3)$$

$$res[1][2] = a[1][0] * b[0][2] + a[1][1] * b[1][2] + a[1][2] * b[2][2]$$

$$= \sum_{k=0}^2 a[1][k] * b[k][2]$$

Homework: Multiply 2 matrices of order $m \times n$ & $p \times q$ where $(n == p)$

1	2
3	4
5	6

3 × 2

×

1	2	3	4
5	6	7	8

2 × 4

=

3 × 4